Tier 3 | Control and safety

Modules 10-14 | SDK 1.62 | 2026-10-29

Mental model

The host owns permissions, deadlines and approvals. The model proposes actions. Guardrails filter content; they do not replace identity or business rules.

Streaming and async

Use stream_async for incremental events and invoke_async for a result. Await I/O; bound concurrency. In notebooks use await instead of nested asyncio.run.

Hooks and interrupts

Hooks observe or enforce lifecycle checks. Interrupts return control for a human decision. Resume only after the host validates the decision and its caller.

Approval checklist

Bind approval to user, action, amount and request ID. Reject stale or mismatched decisions. Use an idempotency key so retries cannot pay twice.

MCP lifecycle

Start client -> discover allowed tools -> invoke agent inside context -> close client. Validate schemas, credentials and transport. Server descriptions are untrusted input.

Prompt injection

Separate retrieved content from instructions. Restrict tool capabilities, validate output, redact telemetry, and test malicious documents. Do not rely only on a prompt.

Useful commands

uv run pytest -q tests/test_tier3_boundaries.py
uv run python scripts/find_leftovers.py --regions us-east-1

Checkpoint

Demonstrate approval denial and safe retry, authenticated local MCP, and a failed injection attempt. Stop local servers and delete optional Guardrail resources.